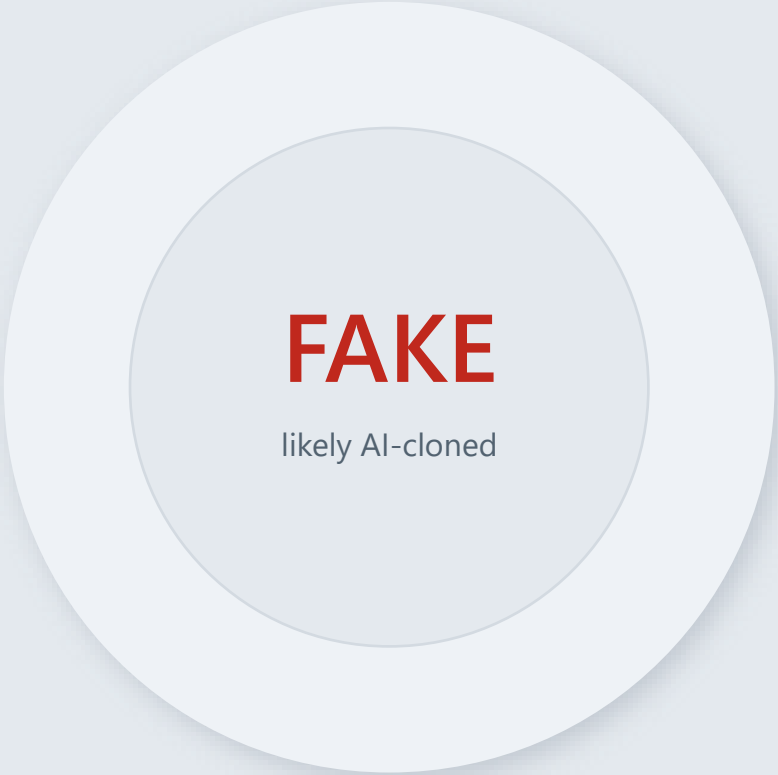
HYPNEXIS'26 · TRACK 01 · AI-01

Kavach

Catching AI-cloned voices before the money moves

Voice-clone detection that keeps working on cloning tools it has never seen



THE PROBLEM

Seconds of your voice are enough to rob your family

1 Collect

A reel, a WhatsApp voice note or a short "wrong number" call gives the scammer 3–10 s of audio.

2 Clone

Free open-source tools copy the voice in minutes.

3 Call

"Papa, accident ho gaya, jaldi paise bhejo." Same voice, real panic.

4 Pay

Money leaves by instant transfer before anyone checks.

Why it works

People trust a familiar voice, and today's phones give no signal that a voice might be synthetic.

WHY DETECTION FAILS

Most detectors learn the wrong thing

Unseen cloners

Trained on known tools, they memorise those tools' quirks and miss new ones.

Shortcuts

Benchmarks leak the answer through silence length and loudness. Models learn that instead of the voice.

Short clips

2–10 s carries little evidence.

Many languages

Training data is mostly English. Scam calls in India are not.

OUR INSIGHT

Every cloner ends in a vocoder

Text + target voice

what to say, in whose voice

→

Cloning model

changes with every new tool

→

Vocoder / codec

turns it into sound: leaves fingerprints

So we detect the fingerprint, not the tool

XLS-R, a speech model pretrained on 436,000 hours of real speech in 128 languages, is kept frozen. It already knows what real speech looks like at every level. We read all 25 of its layers, learn which ones carry the fake signal, and train only a small classifier on top. A frozen model cannot overfit to one cloner's artifacts.

HOW IT WORKS

From a 2–10 second clip to a verdict in under a second

Clean
trim silence
equal loudness

→

Phone
8 kHz line,
codec, noise

→

Ears
XLS-R 300M
frozen, 25
layers

→

Mixer
learned layer
weights

→

Judge
small MLP
score 0–100

→

Verdict
REAL / NOT
SURE
/ FAKE

REAL

This sounds like a real person.

NOT SURE

Can't be sure. Verify another way.

FAKE

Likely AI-cloned. Hang up, call back on a saved number.

BEATING THE SHORTCUTS

Proving the model listens to the voice, not the silence

Shortcut-only classifier

Uses only duration, silence and loudness.
EER on unseen speakers: 47.0%

Our score vs shortcut features

Correlation (0 = unrelated):
duration: 0.17
total silence: -0.25
leading silence: -0.27
loudness db: -0.34

What we do about it

Every clip is silence-trimmed and loudness-normalised before the model sees it, so neither carries any information. Training also sees phone-line versions (8 kHz, µ-law codec, noise) so call audio does not confuse it.

RESULTS

Tested only on speakers and cloners it never saw

Test	What it proves	EER	n
In-the-Wild, unseen speakers, 4 s	Works on people it never heard	2.3%	4026
Same, 2 s clips	Very short evidence	2.6%	4026
Same, 10 s clips	More evidence	2.3%	4026
Same, phone-line audio	Survives a phone call	2.9%	4026
Unseen cloner: KOKORO	Never in training	0.1%	46
Unseen cloner: RVC	Never in training	1.0%	97
Unseen cloner: SNAC	Never in training	8.7%	300

EER = equal error rate: real voices called fake = fake voices called real. Lower is better; 50% is a coin toss. Every number is produced by scripts in the repo (runs/*.json).

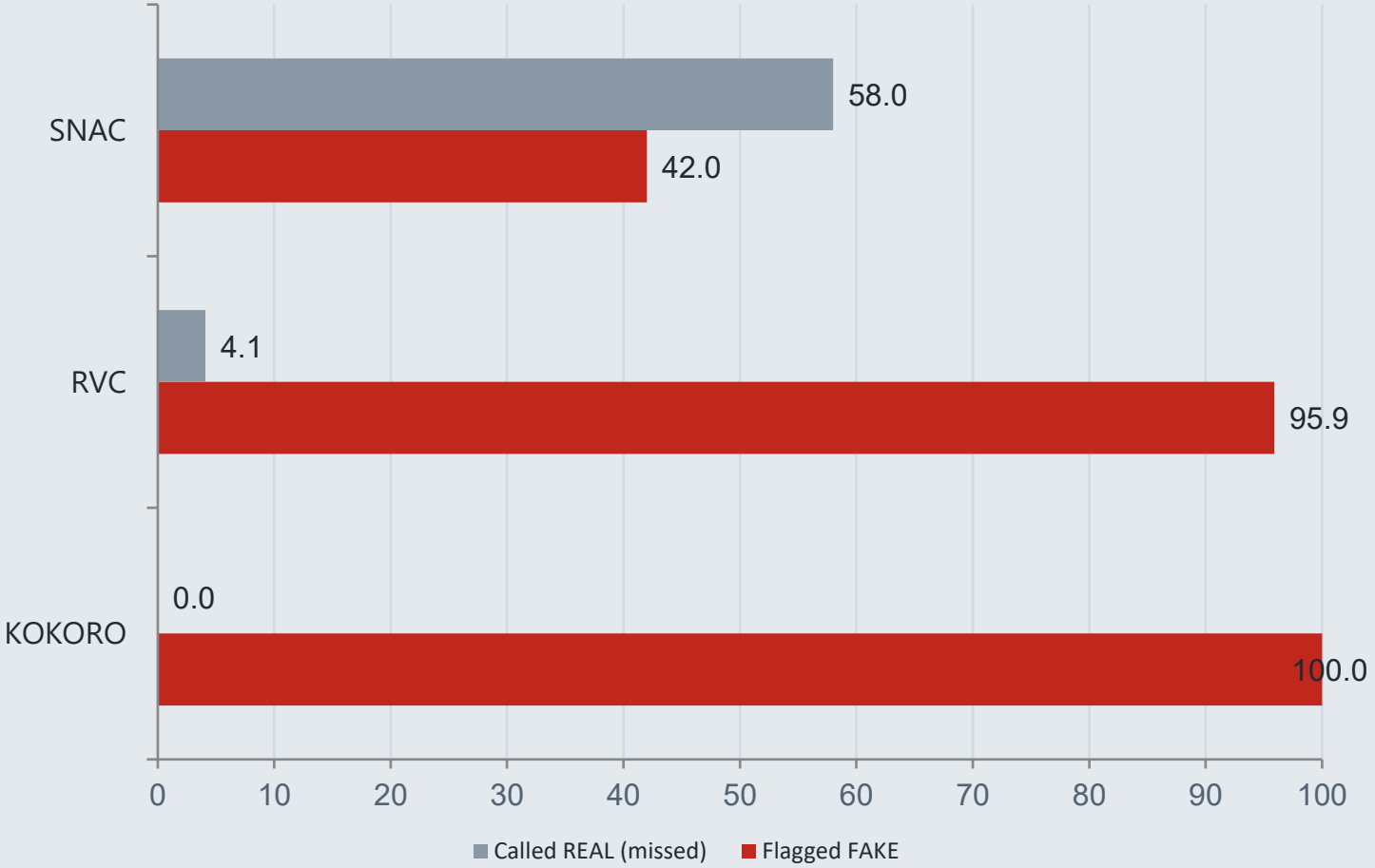
CLOSING THE MODERN-VOICE GAP

Modern AI voices and replayed audio

Test (never seen in training)	Why it matters	EER	Flagged FAKE	n fake
Unseen modern TTS voices (Microsoft neural, Kokoro)	GPT / Gemini-class quality	0.5%	98.0%	204
Same voices played through a speaker into a mic	How a scam call is overheard	10.3%	93.1%	204
In-the-Wild, unseen speakers	Internet deepfakes	2.0%	97.1%	1687
Clean real speech (LibriSpeech)	Real voices must not be flagged	–	false alarm 0.0%	450 real
Clean real speech, replayed	Real voices must not be flagged	–	false alarm 18.9%	450 real

First model (In-the-Wild only) called GPT/Gemini voices REAL. We added modern neural TTS and simulated speaker-to-mic replay to training, holding out 25% of voices for this test.

Share of clones flagged FAKE, by cloning tool



The four tools

SNAC: real voice passed through a neural codec (vocoder fingerprint only)

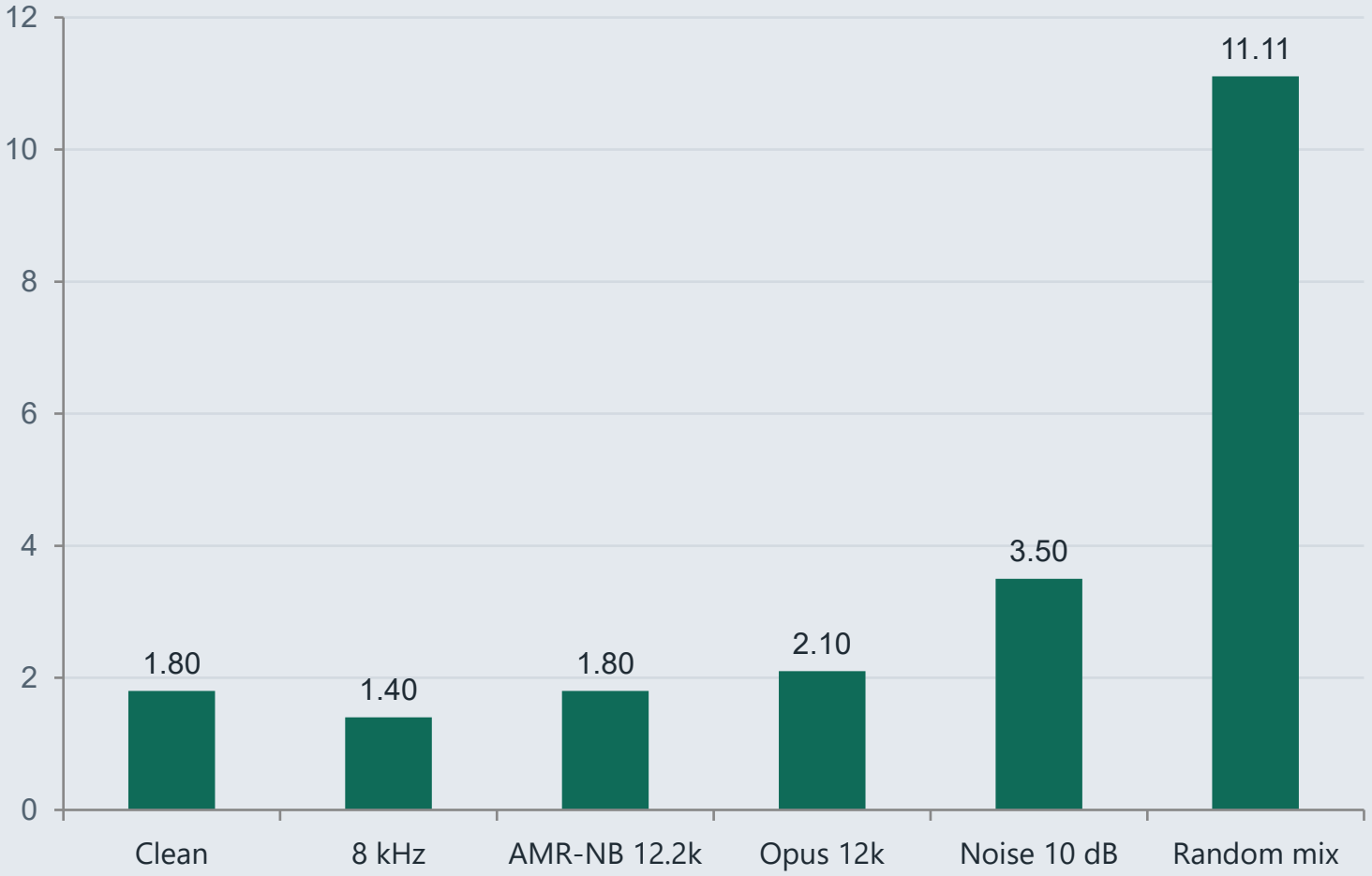
Kokoro: open TTS, English + Hindi voices

Veena: Indian Hindi/Hinglish TTS

RVC: voice conversion into a team member's voice (with consent)

PHASE 2 · R1

Survives phone lines, codecs and noise



Target met

Worst case: noise = 1.94× clean EER

Target ≤ 2.5×

How

Real encoders via ffmpeg (AMR-NB = mobile calls, Opus = WhatsApp/VoIP). Training sees random channels on real and fake alike.

PHASE 2 · R2

Finding the faked seconds inside a real call

F1 80.5%

at 0.5 s resolution on 100 partial fakes

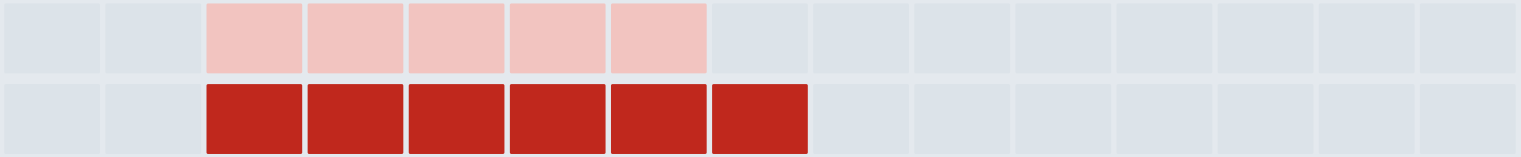
precision 79.9% · recall 81.0%

Partial fakes

Real clip of a speaker with 1–3 s replaced by a fake of the same speaker, loudness-matched, 20 ms crossfade. Threshold set on 60 separate calibration fakes.

Example 1: true fake 3.5–6.2 s

Example 2: true fake 1.0–3.6 s



Example 3: true fake 3.1–5.8 s

Top strip = true fake cells · bottom strip = Kavach's 0.5 s cells

PHASE 2 · R3 + R4

Live call mode, and Phase 1 still holds

Streaming every 500 ms

Browser streams the call over a WebSocket. Every 0.5 s the latest 2 s is scored.

Processing per update: median 25.7 ms, p95 31.4 ms (cuda)

In the app

Live call: mic or simulated call, score line updating twice a second, alert after the voice looks cloned.

Phase 1 test	Phase 1	Phase 2	Change
Unseen speakers 4 s	2.3%	2.0%	-0.30 pts
2 s clips	2.6%	2.3%	-0.30 pts
10 s clips	2.3%	2.0%	-0.25 pts
Phone line	2.9%	3.0%	+0.10 pts

All Phase 1 EERs within 2 points ✓

RED-TEAM TEST

A 4-second WhatsApp note was enough to clone a voice

The attack

Zero-shot cloner F5-TTS (never in training) copied a speaker from a 3.9 s WhatsApp voice note, then we laundered the clones: WhatsApp Opus, phone AMR, street noise, speaker-to-mic replay.

	FAKE	NOT SURE	REAL (missed)
Clones · before noise shield	12	0	6
Clones · with noise shield	16	2	0
Real voice note · with shield	0	1	5

Noise shield on 300 held-out clips

Noisy fakes (15 dB): caught 76% (was 54%), NOT SURE 10%. Noisy real voices wrongly FAKE: 2%. Clean real voices wrongly FAKE: 1%.

How: when a clip is noisy (estimated SNR < 24 dB), Kavach also scores it after removing background noise with Demucs. A red FAKE needs a strong denoised score; a weaker one gives amber NOT SURE instead of a red accusation.

LIVE PROTOTYPE

Upload or record, verdict in under a second



Speed (RTX 4050 laptop)

Median 318 ms per clip, p95 728 ms; 0 errors in 60 concurrent requests.

Demo

1. Team member's real voice → REAL
2. Same person, cloned: "Mummy, I'm in trouble, send ₹50,000" → FAKE
3. A cloner it never saw → still FAKE
4. A judge speaks into the mic → REAL

BUILT TO BE TRUSTED

Safety, privacy and testing

Safety

NOT SURE instead of guessing

Advice always says: verify by calling back

Thresholds set on a separate calibration split

Privacy

Audio analysed in memory, never stored

No audio in logs

Every cloned voice belongs to a consenting team member

Tested

53 unit, API, streaming and security tests: all pass

Browser end-to-end: 13/13

Semgrep (226 rules) and Bandit: 0 open findings

Upload caps, decode timeout, origin-checked WebSocket, strict CSP

IMPACT

Where Kavach goes next

Banks and call centres

Flag cloned voices impersonating customers.

Telecom networks

Warn on suspicious calls at network level.

Families

Phone app, or a ₹1,300 Bluetooth guard for parents (apps cannot read call audio).

Next

Live call mode, more Indian languages, combine with scam-phrase detection.